

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI – 590 018, KARNATAKA, INDIA



**INTERNSHIP REPORT ON
“LAPTOP PRICE PREDICTION AND
RECOMMENDATION”**

Industry Internship

**Submitted in partial fulfillment of the requirements for the award of
BACHELOR OF ENGINEERING**

in

**ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING**

Submitted By

Name
Krithika D P
Rashmi R
Smruthi U
Yogyashree K

USN
4VP22AI028
4VP22AI045
4VP22AI055
4VP22AI061



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha, Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka

Nehru Nagara, Puttur – 574 203, DK, Karnataka, India

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha, Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka

Nehru Nagara, Puttur – 574 203, DK, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



CERTIFICATE

Certified that the Internship, carried out by **Ms. KRITHIKA D P** bearing USN **4VP22AI028**, **Ms. RASHMI R** bearing USN **4VP22AI045**, **Ms. SMRUTHI U** bearing USN **4VP22AI055**, **Ms. YOGYASHREE K** bearing USN **4VP22AI061** a bonafide student of **Vivekananda College of Engineering & Technology, Puttur**, in partial fulfillment for the award of **Bachelor of Engineering** in **Artificial Intelligence & Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2025 – 26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library.

The Internship report has been approved as it satisfies the academic requirements in respect of Industry Internship (21INT82) prescribed for the said Degree.

Internship Coordinator

HOD

Principal

Prof. Venkatesh Y C

Prof. Abhishek Kumar K

Dr. Mahesh Prasanna K

ACKNOWLEDGEMENTS

We undertook a internship at **Prinston Smart Engineers, Bengaluru**, in the domain of **Data Science**. We sincerely express our gratitude to Prinston Smart Engineers, Bengaluru, for providing us with the opportunity to undertake an internship at their esteemed organization. The experience has been immensely valuable, allowing us to enhance our technical skills and gain practical industry exposure. We are especially thankful to the entire team for their continuous support, guidance, and willingness to share their expertise. Their mentorship has greatly contributed to our learning and professional growth.

We would like to extend my heartfelt gratitude to our internship coordinator, **Prof. Venkatesh Y C**, Assistant Professor, Department of AIML, for their exceptional guidance, unwavering support, and motivation throughout the Internship.

We would like to convey my heartfelt gratitude to **Prof. Abhishek Kumar K**, Head of the Department of Artificial Intelligence & Machine Learning, for his invaluable guidance and inspiration.

We sincerely thank our Principal, **Dr. Mahesh Prasanna K**, for offering all the required facilities and fostering a supportive environment to complete the Internship.

We extend our heartfelt thanks to the management, teaching, and non-teaching staff of the Artificial Intelligence and Machine Learning department for their invaluable support and assistance throughout the Internship.

ABSTRACT

This project employs a suite of Python libraries for comprehensive data analysis and machine learning tasks. Utilizing NumPy and Pandas for data preprocessing, Matplotlib for visualization, and Scikit-learn for modeling, the study develops a Laptop Price Prediction and Recommendation System using the Random Forest Regression algorithm. The system is trained on a real-world laptop dataset containing features such as processor type, RAM, storage capacity, storage type, operating system, display size, and user ratings to accurately predict laptop prices in Indian Rupees.

The focus is on efficient model evaluation using metrics like R^2 Score and Mean Absolute Error (MAE), achieving an R^2 score of above 0.85, which demonstrates strong predictive accuracy. In addition to price prediction, the system incorporates a personalized recommendation engine that suggests the top five laptops based on user preferences such as budget, use case (gaming, student, office), minimum RAM, and preferred operating system. The backend is built using the Flask framework and deployed publicly using ngrok, while the frontend is developed as a responsive HTML interface named LaptopIQ. This project establishes a robust and user-friendly foundation, showcasing the effectiveness of machine learning tools in addressing real-world laptop price estimation and recommendation challenges.

TABLE OF CONTENTS

Title	Page No.
List of Figures	I
CHAPTER 1 INTRODUCTION	01
1.1 About the Company	02
1.2 Company Profile	02
CHAPTER 2 INTERNSHIP PROJECT OVERVIEW	03
2.1 Introduction to the Project Undertaken During the Internship	04
2.2 System Requirements	04
2.2.1 Functional Requirements	05
2.2.2 Non Functional Requirements	06
2.3 Software Requirements	06
2.4 Design	07
2.4.1 Flowchart	07
2.4.2 UML Diagram	09
2.4.3 Sequence Diagram	10
CHAPTER 3 INTERNSHIP PROJECT IMPLEMENTATION	11
3.1 Introduction	12
3.2 Algorithm Used	12
3.2.1 Workflow	14
3.3 Pseudo Codes	16
3.4 Results	18
CHAPTER 4 ASSESSMENT & CONCLUSION	20
4.1 Assessment of Internship	21
4.2 Conclusion	21
REFERENCES	23

LIST OF FIGURES

Figure No.	Title	Page No.
2.1	Flowchart of the project	8
2.2	UML Diagram	9
2.3	Sequence Diagram	10
3.1	Laptop Price Prediction Interface	18
3.2	Laptop Price Prediction Result	19
3.3	Laptop Recommendation Result	19

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 About the Company

Prinston Smart Engineers is a Bangaluru-based engineering and IT services company established in 2016, specializing in electro-mechanical solutions and skill development training. The company provides end-to-end services including design, construction, installation, upgrading, and maintenance of electro-mechanical systems such as electrical infrastructure, HVAC, and plumbing. Its scope of work also includes engineering documentation, submittals, shop drawings, coordination drawings, commissioning, start-up, and preparation of as-built drawings in compliance with consultant specifications.

In addition to engineering services, the company offers IT solutions such as web development, cyber security, and software services, along with live instructor-led online training programs. It serves students and professionals across India by collaborating with more than 400 engineering colleges and providing internship opportunities and skill-based learning. The organization focuses on bridging the gap between academic education and industry requirements by offering affordable and accessible training programs, especially in response to the high number of engineering graduates lacking practical skills.

Overall, the company integrates engineering expertise with modern IT services and training initiatives, aiming to enhance employability, support industry needs, and provide practical exposure through real-time projects and professional guidance.

1.2 Company Profile

Company Name : Prinston Smart Engineers

Main Office : Anthony Building 2nd Floor, Begur Road, Above Manya Dental Clinic, Begur, Bangaluru, Karnataka 560068

Email : info@prinstonsmart.com , prinston.smart@gmail.com

Year Stand Up: 2016

Company Category: Engineering, IT Services, and Training Company

INTERNSHIP PROJECT OVERVIEW

CHAPTER 2

INTERNSHIP PROJECT OVERVIEW

2.1 Introduction to the Project Undertaken During the Internship

The Laptop Price Prediction and Recommendation System is a machine learning-based web application developed during the internship at Princeton Smart Engineers. The system enables users to predict the price of a laptop based on key hardware specifications and also recommends the best laptops within a given budget and use-case preference. The project was built using Python in Google Colab, leveraging the Random Forest Regressor algorithm for price prediction and a custom scoring algorithm for intelligent recommendations. A Flask REST API serves as the backend, and the frontend is a responsive HTML-based web interface that communicates with the API over the internet using ngrok tunneling.

2.2 System Requirements

The system requirements represent the necessary functionalities and requirements for an effective operation of the Laptop Price Prediction and Recommendation System. The system requirements determine the main tasks and functionality of the proposed solution. In order to organize the requirements clearly, both functional and non-functional requirements are presented.

Functional requirements determine specific functionalities associated with the processing of the laptop price prediction task. Namely, the functional requirements of the application involve collecting users' information about their desired laptops, processing this information, calculating the price of the requested laptop using a trained machine learning model, and providing predictions via a web interface. Also, functional requirements include connecting the front-end and back-end of the application via the Flask API.

In turn, the non-functional requirements cover performance and accuracy, usability, efficiency, and reliability of the system. The predicted prices have to be provided promptly; the application has to provide stable performance and work in real time. Furthermore, the system should provide users with convenient interaction possibilities.

Thus, the discussed requirements guarantee efficient performance and accurate predictions from the application.

2.2.1 Functional Requirements

Functional Requirements describe how the Laptop Price Prediction and Recommendation System works with regard to accepting input from the user, processing it and giving the output.

- **User Input Specification:**

The software will accept as input the specifications of laptops such as RAM, storage, processor, operating system, screen size, and ratings.

- **Price Prediction:**

This is where the given inputs will be used in order to predict the price of the laptop that will be given in the form of Indian Rupees using Machine Learning Model.

- **Recommendations Generation:**

It will recommend the top five laptops to users based on their budget and use-case (e.g. Gaming, student, office and general.)

- **Dataset Processing:**

The real-time laptop dataset will be loaded by the system from Google Drive.

- **REST API Services:**

The system will include REST API services such as /predict, /recommend, and /stats to process requests related to predictions, recommendations, and other system-related information.

- **Presentation of Results:**

The system will present results related to predictions and recommendations in terms of information about laptops like their names, prices, processors, RAMs, storages, operating systems, and display sizes.

- **Preprocessing and Integration of Data:**

The system will preprocess input data from users and integrate it into the trained model.

- **Instantaneous Response:**

The system will respond to all requests instantaneously and effectively for predictions and recommendations.

2.2.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and performance standards of the Laptop Price Prediction and Recommendation System, ensuring that the system operates efficiently and provides a good user experience.

- **Performance:**

The system should return a price prediction within 2 seconds to ensure quick response time.

- **Accuracy:**

The machine learning model (Random Forest) should achieve an R^2 score above 0.85, ensuring reliable and consistent predictions.

- **Scalability:**

The Flask API should be capable of handling multiple concurrent requests through ngrok tunneling without performance degradation.

- **Usability:**

The web interface should be simple, intuitive, and user-friendly, with clearly labeled input fields and output sections.

- **Portability:**

The entire project should run smoothly on Google Colab without requiring local installation or complex setup.

- **Maintainability:**

The code should be well-structured and modular, with separate sections for data preprocessing, model training, API development, and frontend integration, making it easy to update and manage.

2.3 Software Requirements

The following software tools and libraries were used in the development of the Laptop Price Prediction and Recommendation System:

- Programming Language: Python 3.x
- Development Environment: Google Colab
- Machine Learning Library: scikit-learn (RandomForestRegressor, LabelEncoder, train_test_split)
- Data Processing Libraries: pandas, numpy
- Data Visualization: matplotlib
- Web Framework: Flask, Flask-CORS

- API Tunneling: pyngrok (ngrok)
- Model Serialization: pickle
- Frontend Technologies: HTML5, CSS3, JavaScript
- Dataset: laptops.csv (stored in Google Drive)

2.4 Design

The architecture of the Laptop Price Prediction and Recommendation System has been devised to promote simplicity, efficiency, and maintainability. The design adheres to the principle of modularity, whereby the front end, back end, and machine learning algorithms have been divided into separate modules. The design promotes effective management of the system and facilitates its enhancement in the future. While the frontend deals with the presentation of data to the user, the back end processes the data and interacts with the trained machine learning algorithm.

The system has been built with an intuitive and user-friendly graphical user interface. It enables the user to provide information about the RAM, processor, storage, and other technical features of the laptop without difficulty. Appropriate layout techniques have been used to present the fields for input and output of information to the user.

On the backend level, the system accepts and handles all requests submitted by the frontend with the help of a Flask API. The provided data is pre-processed before being used to obtain predictions by the trained machine learning model. The communication of all components is facilitated by calling APIs. Ngrok can be used for external access to the application. Simple input validation methods will be used to handle any potential incorrect user data. Error-handling will be added to manage any possible exceptions during the processing. The architecture is scalable and can be easily modified in the future with respect to any possible additions or improvements.

2.4.1 Flowchart

Designing the flowchart for the Laptop Price Prediction and Recommendation System involves creating a clear visual representation of how the system works and how different components interact with each other. It outlines the step-by-step flow of user actions. This helps in understanding the overall structure and behaviour of the application.

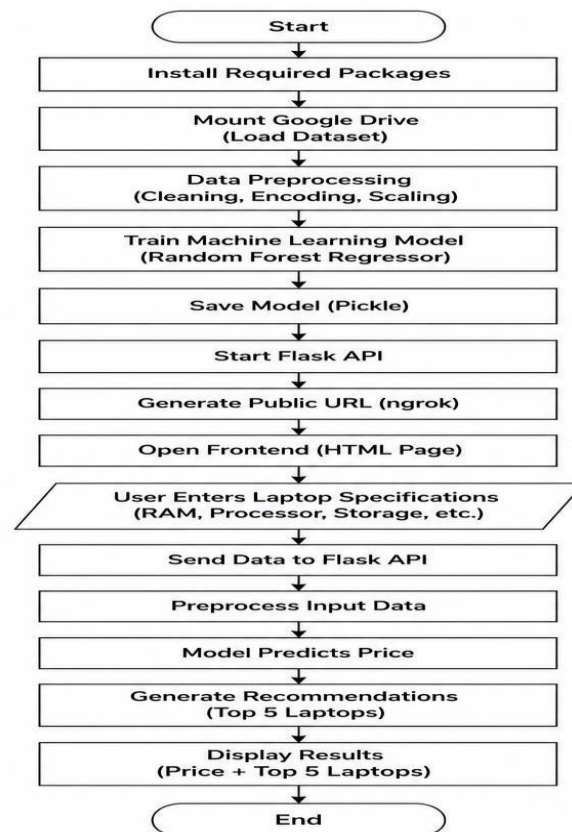


Figure 2.1: Flowchart of the project

Figure 2.1 below represents the flowchart for the Laptop Price Prediction and Recommendation System. First, the installation of the necessary Python libraries is done, followed by mounting Google Drive to read the dataset. After loading the dataset, the next step is to preprocess the data by cleaning, encoding, and scaling the data before training the machine learning model. The model will learn from the dataset and be trained and finally saved for future use.

The Flask API will be started to receive user requests and use ngrok to get a public URL to allow users to access the system from outside. Once the system is ready, the user will visit the HTML page through the browser and enter laptop information like RAM, processor, storage, and others. This data will be preprocessed and passed back to the Flask API.

After that, the model will calculate the price of the laptop, and finally, the computer will recommend the best five laptops that satisfy the user's criteria. Both the estimated price and the laptop recommendations will be shown to the user. Thus, the operation comes to an end.

2.4.2 UML Diagram

UML diagram is drawn to show how the laptop price prediction and recommendation system can be implemented. These diagrams can provide a lot of information about interaction of system components, including frontend and backend processes and machine learning algorithm. Different types of diagrams, such as use case, class, and sequence diagrams, can be used to depict user interaction with the system, system structure, and data flow within the system. Specifically, the use case diagram can describe how users enter data about specifications and get predictions, while the sequence diagram can describe interaction of system components with each other.

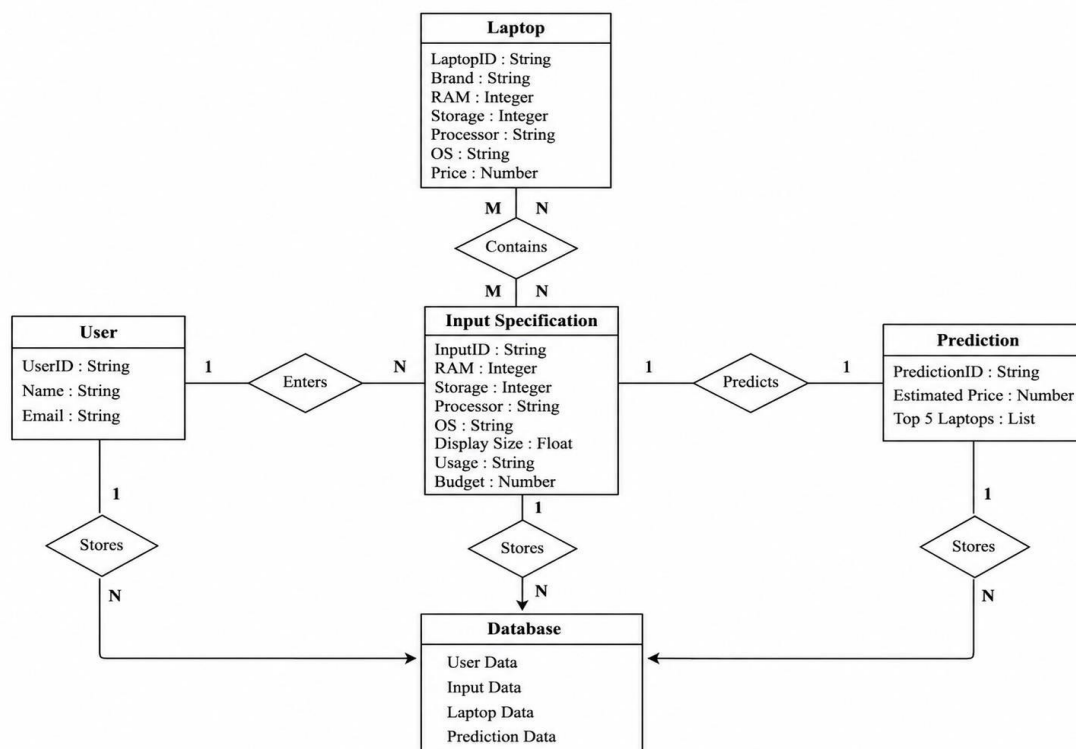


Figure 2.2: UML diagram for the system.

The figure 2.2 demonstrates UML diagram of the Laptop Price Prediction and Recommendation System where the user entity consists of details regarding the user ID, user name, and email address and interacts with the system by giving specifications for the laptop. The laptop entity will include details such as laptop brand, RAM, storage, processor, operating system, and laptop price and is related with input specification entity in a many-to-many relationship indicating that there could be more than one laptop compared according to different inputs.

Thus, the input specification entity becomes an intermediary entity consisting of details such as RAM, Storage, Processor, Operating system, screen size, purpose of laptop usage

and budget of user and then through "predicts" relationship changes itself into a prediction entity consisting of prediction price and top five recommendations of laptops. Thus, it becomes the most significant functional part of the whole process.

Continuous interaction of the user entity keeps interaction between user and the system going on. Finally, user, input specification, laptop and prediction entity are related with database containing data about them respectively.

2.4.3 Sequence Diagram

A sequence diagram shows how components of the **Laptop Price Prediction and Recommendation System** interact step by step. The user enters specifications, which are sent to the Flask API. The backend preprocesses data and passes it to the model. The model returns predicted price and recommendations, which are displayed to the user.

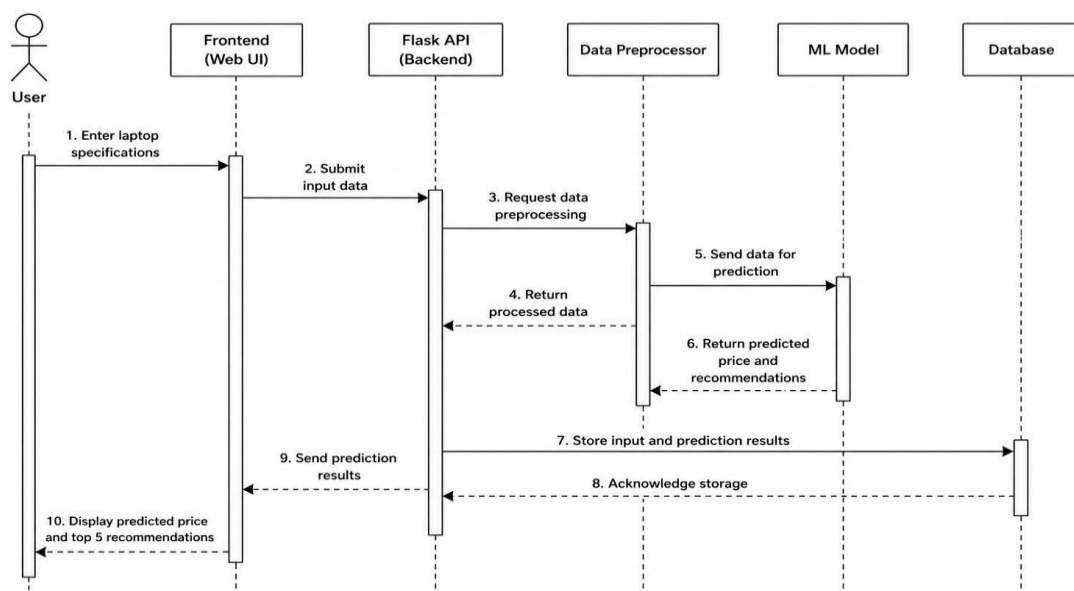


Figure 2.3: Sequence diagram for the system.

Figure 2.3 shows the sequence diagram of the **Laptop Price Prediction and Recommendation System**, where the user begins by entering laptop specifications through the frontend interface. The frontend sends this data to the Flask API, which preprocesses the input and forwards it to the machine learning model. The model processes the data and returns the predicted price along with recommendations to the backend. The backend then sends the results back to the frontend, where they are displayed to the user. This sequence illustrates the step-by-step interaction between components to complete the prediction process efficiently.

INTERNSHIP PROJECT IMPLEMENTATION

CHAPTER 3

INTERNSHIP PROJECT IMPLEMENTATION

3.1 Introduction

Development of a data science application of predicting and recommending laptop prices using machine learning algorithms forms the objective of implementing the laptop price prediction and recommendation system. The process involves application of data science techniques including data preprocessing, feature engineering, modeling, and prediction analysis to make the system a real-life application. For the purposes of model development and predictions, Google Colab will be utilized. Furthermore, Flask framework will be used as the backend API while a simple HTML file will form the frontend interface for user entry and results presentation. Users will provide input concerning the laptop specifications for predictions and recommendations.

Data science techniques for developing the data science application include pre-processing and engineering of the laptop dataset. The steps involve removal of missing values from the dataset as well as extraction of relevant features such as memory capacity, storage capacity, type of processor, and type of OS to engineer the laptop model. To develop a model capable of predicting the laptop price accurately, the Random Forest Regression algorithm is implemented. The process is complemented by recommending laptops to users in line with their preferences for budgets, minimum RAM size, usage, and OS. Various error handling and validation techniques have been used to guarantee reliable system performance and processing of user inputs properly. Ngrok has been used for deploying the software such that its backend becomes accessible via a public website. The end result has been an efficient and smart laptop price prediction and recommendation tool.

3.2 Algorithm Used

Laptop Price Prediction and Recommendation System employs both machine learning techniques and data processing algorithms for carrying out operations such as price prediction, laptop recommendation, and efficient user interaction. These algorithms assist in data preprocessing, model training, predictions, laptop filtering, and recommendation ranking. The important algorithms employed in this project are discussed below:

- **Random Forest Regression Algorithm**

Random Forest Regression is the primary algorithm employed in this model for predicting the price of laptops. Random Forest is a machine learning algorithm designed to build and evaluate various decision trees for better predictive performance and generalization. The algorithm operates by building decision trees using various subsets of data and combining their results to predict the final output. In this case, the algorithm is applied to establish the connections between various features of laptops including RAM, processor type, capacity of storage, operating system, display size, and ratings as well as their prices.

- **Data Preprocessing Algorithm**

The system uses a preprocessing algorithm to clean and transform the raw laptop dataset into a machine-learning-ready format. The preprocessing process includes handling missing values, extracting numerical values from text fields, normalizing operating system names, identifying processor brands, and calculating processor performance tiers. For example, RAM values such as “8 GB” are converted into numerical format, and storage values such as “1 TB SSD” are converted into total storage capacity in GB. This algorithm ensures consistency, improves data quality, and prepares structured input for model training.

- **Label Encoding Algorithm**

In order to deal with the categorical attributes like processor brand and operating system, a label encoding technique is employed. Since all ML models can take input only in numeric form, it becomes necessary to convert categorical data into numeric data. Label encoder gives each category a different numeric value. For instance, categories like processor brands such as Intel, AMD, and Apple will be assigned different numbers.

- **Scoring and Ranking Algorithm**

An individualized scoring algorithm is used to determine the relevance of the laptops based on how relevant they are to the user. Scoring is done based on processor grade, RAM, size of storage space, type of storage space, rating, and proximity to the user's price range. Other factors that influence the ranking of a particular laptop include the use case of the computer (whether for gaming, studies, or office work).

- **Sorting Algorithm**

In order to show the best suggestions, sorting is employed as an algorithm for arranging the laptops according to their score values. Those with high scores will be prioritized in the suggestions provided. This method enables the arrangement of data into an

understandable format where users are able to determine the appropriate laptop choices without making comparisons to all models available.

3.2.1 Workflow

The Laptop Price Prediction & Recommendation System operates through an established workflow that facilitates the process of data processing, price prediction, and recommendation of laptops. This system uses a combination of machine learning, backend API development, and frontend for ease of use by the client. The workflow of this system is detailed as follows:

- **Dataset Loading and Initialization**

In the beginning of the workflow, imported the laptop data from cloud storage through Google Drive which is integrated with Google Colab. The laptop data includes the information about processor, RAM, storage space, operating system, size of screen, rating, and price of laptops. Once the laptop dataset is successfully loaded, we validate the dataset structure and its shape.

- **Data Preprocessing and Cleaning**

Once the loading of the dataset is completed, the system undergoes data preprocessing steps that will help in cleaning and formatting the data for machine learning modeling. Data entries that have missing values in critical columns such as rating, number of reviews, and number of ratings are processed accordingly using relevant data imputation techniques. The specification of the RAM, storage capacity, processing capabilities, and operating systems in texts is also transformed into numeric data. New features are generated by performing operations on existing features; this includes RAM capacity in GB, total storage capacity, storage types score, CPU manufacturer name, and CPU grade.

- **Feature Encoding and Selection**

The categorical variables like processor brand and operating system are encoded into numeric values after the preprocessing phase. This helps the machine learning algorithm interpret and process categorical data appropriately. Some of the key factors, including RAM, storage capacity, storage type, processor brand, processor tier, operating system, screen size, and rating, are considered as input features for the algorithm.

- **Machine Learning and Model Training**

Following the processing of the features, the data set is split into two categories, namely the training set and testing set. The training set is used in training the machine learning

algorithm, while the testing set is used in evaluating the performance of the machine learning algorithm. The Random Forest regression algorithm is used in predicting the price due to its ability to capture complex relationships between laptop features and their prices.

- **Model Evaluation and Analysis**

Performance measures are applied to the machine learning model after its training. They include measures like R^2 Score and Mean Absolute Error (MAE), which measure the accuracy of the model. Other graphical analysis reports include feature importance plots, price distribution plots, and price prediction versus actual plots that indicate how the features affect the price of laptops.

- **Flask API Integration**

The saved machine learning model is integrated into a backend API using Flask. The API acts as the communication layer between the frontend and backend. It provides separate routes for laptop price prediction, laptop recommendation, and system statistics. This allows users to interact with the machine learning model through simple API requests.

- **Laptop Price Prediction Workflow**

The first module in price prediction involves the entry of laptop features like CPU, memory size, storage capacity, storage type, OS, screen size, and rating. These details get captured, and an API request containing the inputs is sent from the frontend interface to Flask API. The inputs go through a processing step, which includes extraction of important features, feature encoding, and creation of the feature vector. Once this is done, the machine learning algorithm predicts the estimated price of the laptop and responds back to the frontend interface.

- **Laptop Recommendation Workflow**

Recommendation Module helps in getting recommendations for laptops according to user's preference. Here, user enters various parameters like budget, usage (for students, gaming, office), RAM capacity required, and OS preference. This module selects various laptops from available database according to the parameters mentioned by users and generates scores for each selected laptop using different parameters, like the processor level, RAM capacity, storage space, OS level rating, and budget range. This score is generated according to certain predefined weightings.

- **Frontend User Interface Interaction**

The frontend of the system is made using a basic HTML interface. There is a section in the interface where the user can make predictions about the price of the product.

Another section deals with recommendations of laptops based on the specifications entered by the user.

- **Public Deployment using ngrok**

The simple HTML-based frontend for this purpose will be integrated with the help of ngrok. This will provide a public URL through which the frontend will be connected with the backend which is running on the Colab server. Thus, access can be obtained without having the frontend hosted locally.

- **API Testing and Validation**

Prior to accessing the frontend, the API endpoints will be tested with HTTP requests in order to check the performance of the prediction and recommendation components.

- **Result Generation and User Decision Support**

The output from the system is presented to the end user. This output includes the price of the laptop and the laptop recommendations. This will assist the user in comparing different laptop models, the cost of the laptop in the market, and in making sound decisions regarding which laptop to purchase, depending on their budgets.

3.3 Pseudo Codes

Pseudo code snippets are simplified, language-independent representations of algorithms that describe the logic and flow of a program. They help in understanding the step-by-step process of solving a problem and provide a clear way to plan functionalities before implementation.

- **Code for Dataset Loading**

```
DATASET_PATH = '/content/drive/MyDrive/laptops.csv'
df = pd.read_csv(DATASET_PATH)
print(df.shape)
print(df.head())
```

- **Code for Data Preprocessing**

```
def preprocess(df):
    df['rating'] = df['rating'].fillna(df['rating'].median())
    df['ram_gb'] = df['ram'].apply(extract_ram)
    df['storage_gb'] = df['storage'].apply(extract_storage)
    df['processor_brand'] = df['processor'].apply(extract_processor_brand)
    df['processor_tier'] = df['processor'].apply(extract_processor_tier)
```

```
df['os_clean'] = df['os'].apply(extract_os)

return df
```

- **Code for Feature Encoding**

```
le_brand = LabelEncoder()

le_os = LabelEncoder()

df['processor_brand_enc'] = le_brand.fit_transform(df['processor_brand'])

df['os_enc'] = le_os.fit_transform(df['os_clean'])
```

- **Code for Model Training**

```
X = df[feature_cols]

y = df['price(in Rs.)']

X_train, X_test, y_train, y_test = train_test_split(X, y)

model = RandomForestRegressor()

model.fit(X_train, y_train)
```

- **Code for Price Prediction API**

```
@app.route("/predict", methods=["POST"])
def predict():
    data = request.json
    features = prepare_features(data)
    predicted_price = model.predict(features)
    return jsonify(predicted_price)
```

- **Code for Laptop Recommendation API**

```
@app.route("/recommend", methods=["POST"])
def recommend():
    filtered = filter_laptops(user_preferences)
    top_laptops = filtered.nlargest(5, "score")
    return jsonify(top_laptops)
```

- **Code for Recommendation Scoring Logic**

```
def score_laptop(row):
    score = row["processor_tier"] * 20
    score += row["ram_gb"] * 5
```

```
score += row["storage_gb"] * 2
score += row["rating"] * 10
return score
const orders = await Order.find({ userId: req.params.userId });
res.json(orders);
});
```

3.4 Results

The results show that the laptop price prediction and recommendation system successfully performs dataset preprocessing, feature extraction, machine learning model training, price prediction, and laptop recommendation, providing accurate predictions and personalized suggestions through a simple and user-friendly interface.

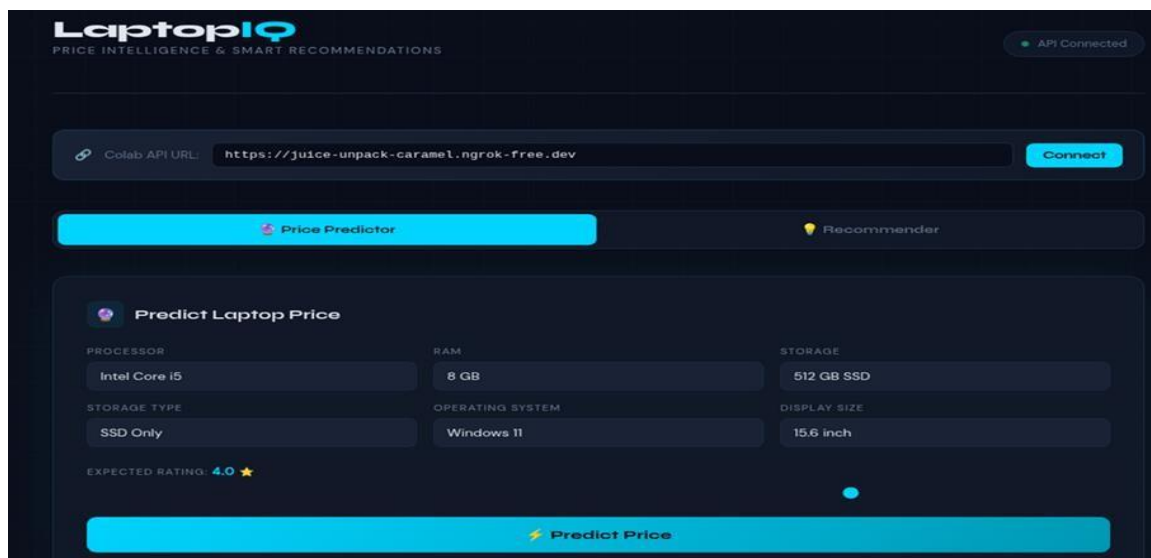
The screenshot displays the LaptopIQ application interface, which is designed for price prediction and recommendations. At the top, the header includes the 'LaptopIQ' logo and the tagline 'PRICE INTELLIGENCE & SMART RECOMMENDATIONS'. A status indicator in the top right corner shows 'API Connected'. Below the header, there is a section for connecting to a Colab API URL, with the example URL 'https://juice-unpack-caramel.ngrok-free.dev' and a 'Connect' button. The main interface is divided into two tabs: 'Price Predictor' (active) and 'Recommender'. The 'Predict Laptop Price' section features a form with six input fields: 'PROCESSOR' (Intel Core i5), 'RAM' (8 GB), 'STORAGE' (512 GB SSD), 'STORAGE TYPE' (SSD Only), 'OPERATING SYSTEM' (Windows 11), and 'DISPLAY SIZE' (15.6 inch). Below these fields, the 'EXPECTED RATING' is shown as 4.0 stars. A large 'Predict Price' button is located at the bottom of the form.

Figure 3.1: Laptop Price Prediction Interface

The following Figure 3.1 depicts the primary interface of the LaptopIQ application. The application offers users an easy-to-use platform where they can predict the laptop price considering various parameters such as processor, memory, storage, type of storage, operating system, size of display, and rating. The interaction between the application and its API takes place using the provided public URL, allowing real-time interaction for prediction and recommendations services. Users simply need to provide laptop specifications and click on the “Predict Price” button to get the predicted laptop price.

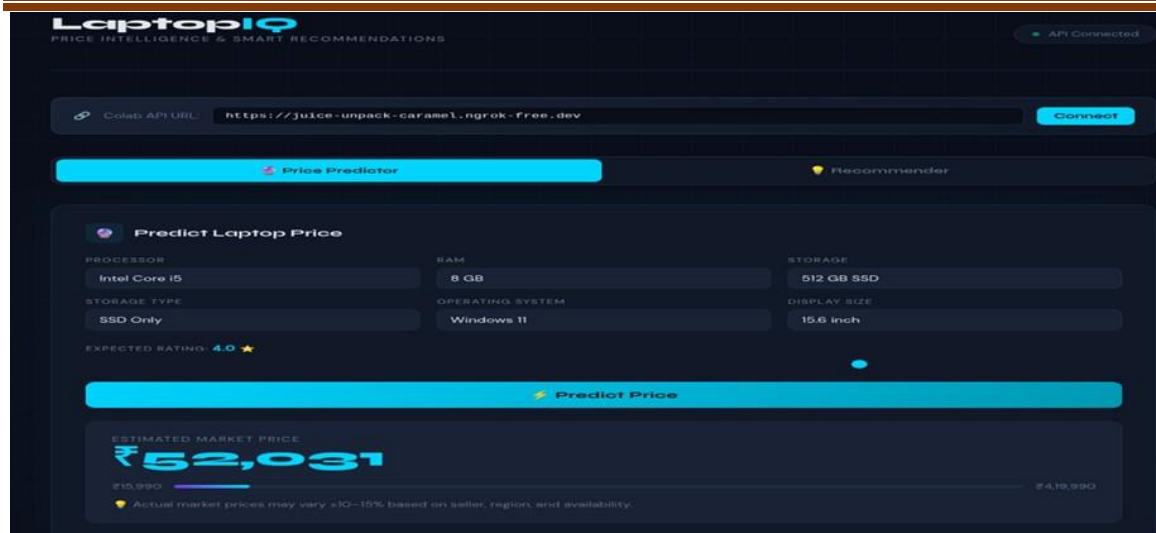


Figure 3.2: Laptop Price Prediction Result

Figure 3.2 illustrates the price prediction result screen of the LaptopIQ system. Once the specifications of the laptop have been inputted such as CPU type, memory, hard disk, hard disk type, OS, screen size, and the rating, the user may simply hit the “Predict Price” button to get the estimated price of the laptop. The system uses the inputted data and runs the machine learning algorithm to give an instant predicted price for the laptop.

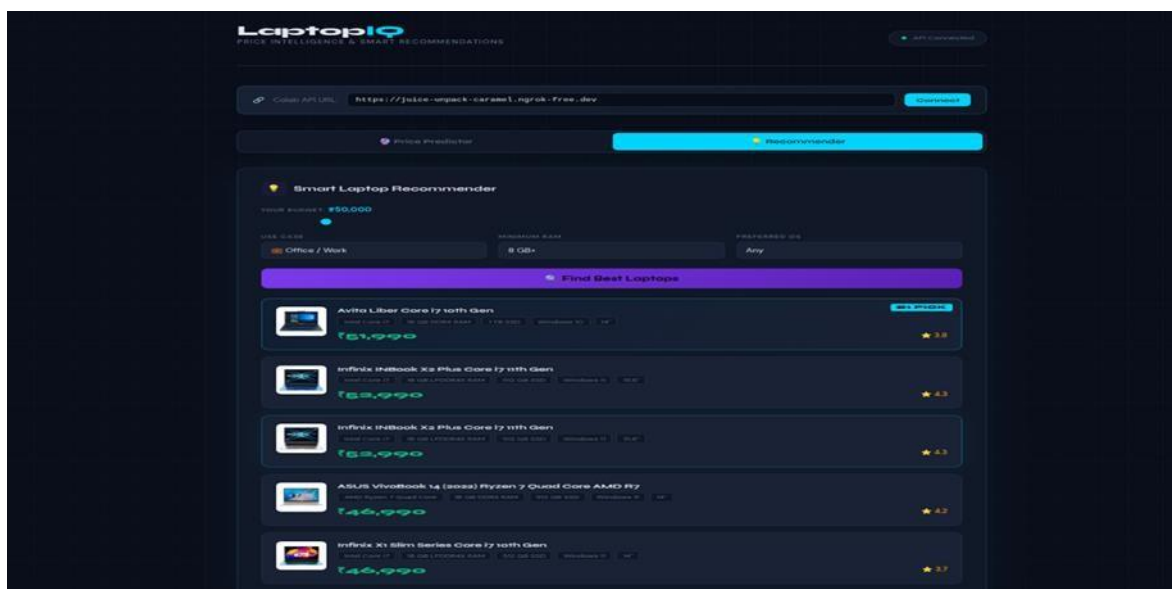


Figure 3.3: Laptop Recommendation Result

The laptop recommendation results page of the LaptopIQ is depicted in Figure 3.3 below. Users can specify their budget, the use case, minimum RAM requirements, and preferred operating system to receive appropriate laptop recommendations. Upon clicking the "Find Best Laptops" button, the system evaluates the available laptops that meet the specified requirements and ranks them accordingly. The list includes the recommended laptops along with relevant specifications, prices, and ratings.

ASSESSMENT & CONCLUSION

CHAPTER 4

ASSESSMENT & CONCLUSION

4.1 Assessment of Internship

I had hands-on experience in the field of data science during my internship, which helped me better understand the practical implementation of machine learning and data-driven applications. Through the Laptop Price Prediction and Recommendation System project, I learned much about data preprocessing, data exploration, feature engineering, building models, and evaluating their performance. The said project enhanced my knowledge about recommendation and prediction system as well as improved my critical and problem-solving skills in applied data science projects. The complete implementation was done in Google Colab, starting from data collection to preprocessing, model training, and API development. The Random Forest Regression model gave an R^2 score of more than 0.85, demonstrating accurate predictions. I also created a recommendation engine that recommends laptops to users based on factors like price, RAM, storage, and CPU.

As a result of the above-mentioned internship, I was able to develop various skills including problem-solving abilities, debugging, and analysis skills. Also, through the use of ngrok tool, I acquired hands-on experience in integrating frontend and backend parts of an application. Finally, some useful tools and technologies that I was able to apply include Python, Pandas, Scikit-Learn, Flask, Git, GitHub, and Postman. Overall, the internship enhanced my technical knowledge, practical skills, and confidence in real-world machine learning application development.

4.2 Conclusion

The internship was a worthwhile learning process through which the theoretical concepts of data science could be put into practice. Developing the Laptop Price Prediction and Recommendation System helped to gain insights into the entire process of machine learning, including data gathering, preprocessing, creating a model, assessing the model, and deploying it in a real-world scenario. The internship helped to develop technical expertise in data preprocessing, designing a machine learning model, analysing the data, and implementing API.

In addition, the internship improved analytical skills and adaptability when dealing with real-world datasets. The insights acquired from the internship will form a solid basis for working on real-world data science and machine learning problems with confidence. Therefore, the internship played an integral part in enhancing practical skills and determining future career prospects in data science.

REFERENCES

- [1] K. Telkar, G. Kandarkar, A. Tola, J. Bagade and R. Bhimanpallewar, “Prediction of Laptop Prices with Recommendations Based on User Specifications”, 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA), pp. 1–6, 2023, doi: 10.1109/ICCUBEA58933.2023.10392000.
- [2] Y. Karakuş and T. T. Bilgin, “Laptop Price Range Prediction with Machine Learning Methods”, International Journal of Multidisciplinary Studies and Innovative Technologies, vol. 8, no. 1, pp. 40–45, 2024, doi: 10.36287/ijmsit.8.1.7.
- [3] G. Anudeep Goud, B. Tanishqi, B. Hemanth Reddy and G. Tanish Bhargav, “Real-Time Laptop Price Prediction through Web Data Analysis”, Proceedings of Advances in Engineering Research, pp. 1–8, 2025, doi: 10.2991/978-94-6463-858-5_115.
- [4] V. Surjuse, S. Lohakare, A. Barapatre and A. Chapke, “Laptop Price Prediction using Machine Learning”, International Journal of Computer Science and Mobile Computing, vol. 11, no. 1, pp. 164–168, Jan. 2022, doi: 10.47760/IJCSMC.2022.V11I01.021.
- [5] A. Acharya, A. Agarwal, A. Agarwal and A. Soni, “A Hybrid Laptop Recommendation System for Engineering Undergraduates”, 2023 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES), pp. 967–973, 2023, doi: 10.1109/CISES58720.2023.10183587.